Name_____ **Microscope Parts and Functions** Do Not Lose!!!!!

Why are microscopes important?

We use a _____. How do we know?

_____

Microscope Part	Function
1. eyepiece	
2. course adjustment	
3. fine adjustment	
4. arm	
5. body tube	
6. nose piece	
7. objective lens	
8. stage	
9. clip	
10. diaphragm	
11. light source	
12. base	

What is the field of view?

How do you calculate the magnification?

How to use a microscope:

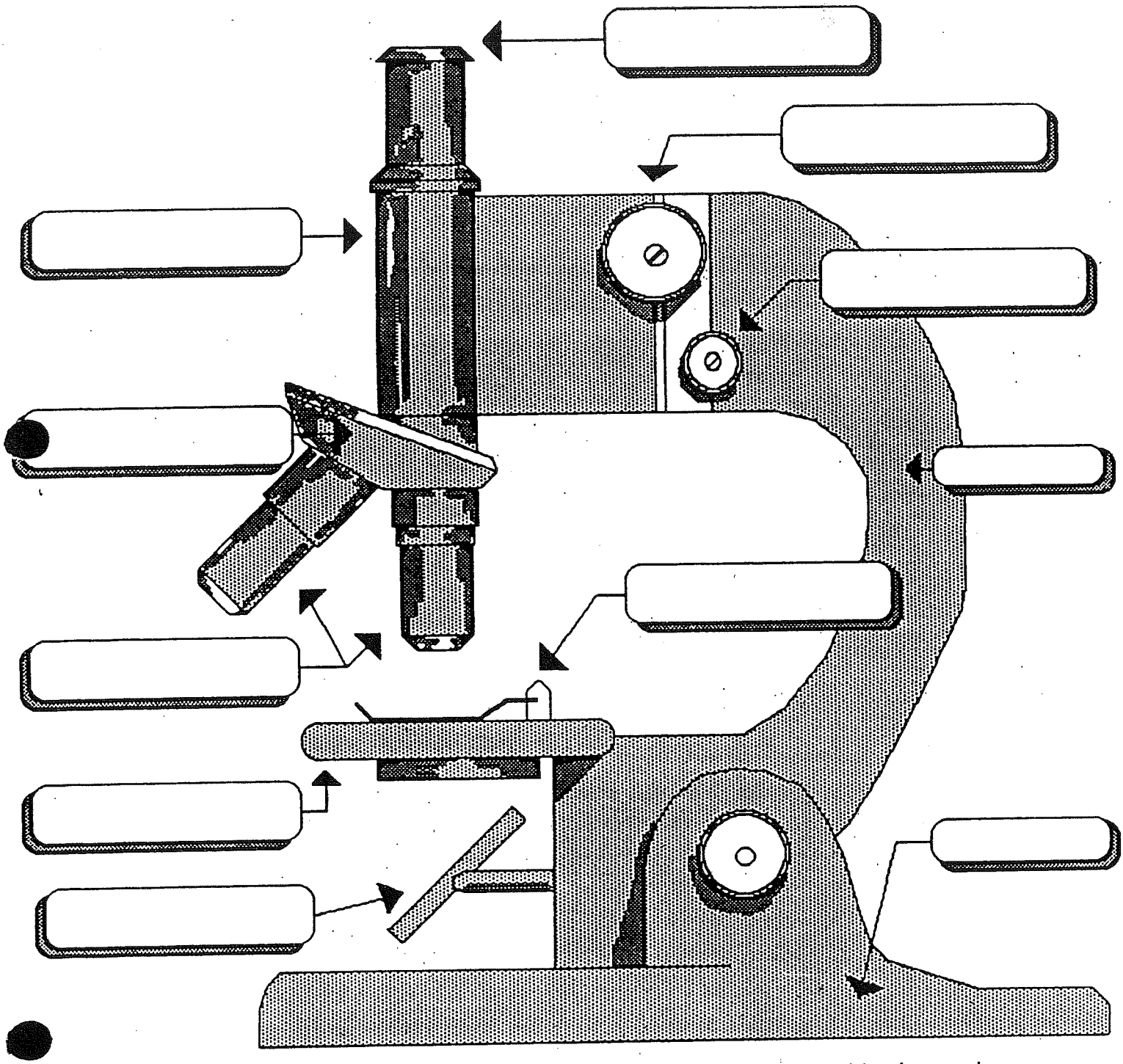
1) Make sure that the stage is completely rolled down before you put a slide on it.
2) Turn the revolving nosepiece to the low power objective lens
3) Look through the eyepiece and turn the course adjustment knob slowly
4) Object should come into focus

http://www.cas.muohio.edu/mbi-ws/microscopes/microscopeparts.html

FINDING THE PARTS OF A MICROSCOPE

Name _____

Directions: Label the diagram of a microscope with the words at the bottom of this page.



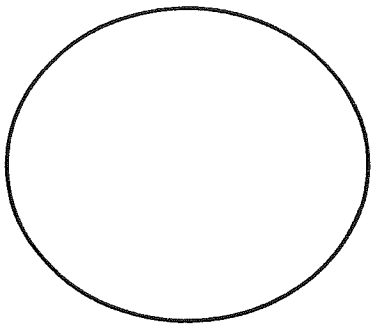
nosepiece mirror fine adjustment knob stage clip tube objectives base

stage arm coarse adjustment knob eyepiece

© 1989 United Learning

NAME:_____ **Period:**_____

Enormous e Lab

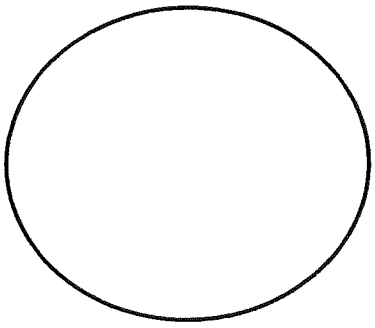


Drawing 1 at _____ magnification

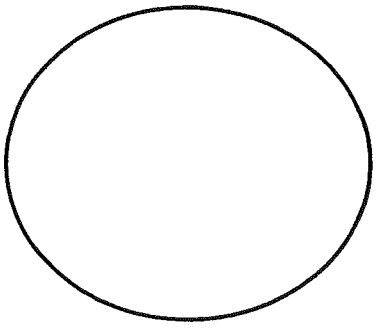
Question 1: While looking at the e through the eyepiece, move the slide slowly to the left. Describe what happened to the e.

Question 2: While looking through the eyepiece, move the slide slowly up. What happened?

Drawing 2 at_____ **magnification**



Drawing 3 at _____**magnification**



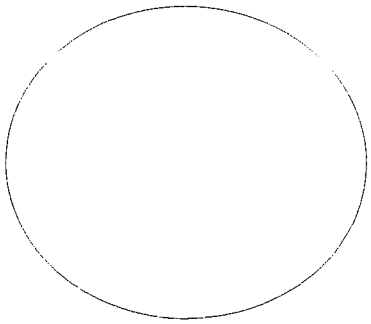
Question 3: How does the movement of the image you see when you look through the microscope compare to the actual movement of the slide?

Question 4: Compare how the e looked under low and high power.

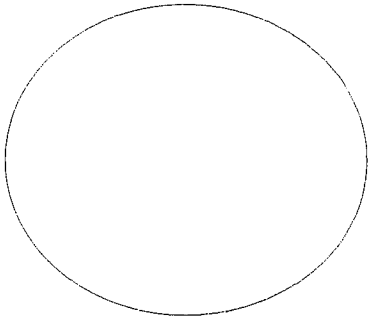
Question 5: What is the "field of view"?

Pick 3 other slides to draw. Be sure to write the power at which you observed the slide and include the name of the slide. Your drawings must have DETAILS.

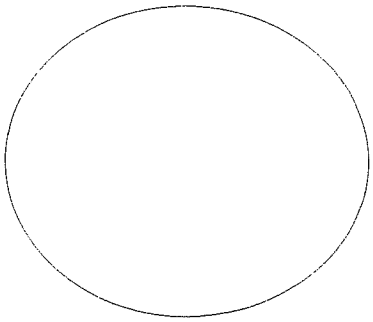
Name of slide_____
Magnification_____



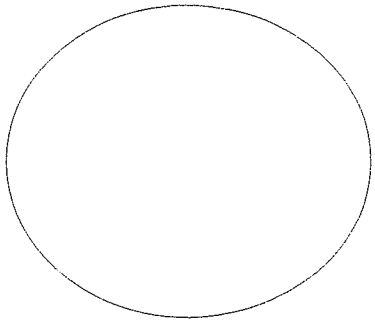
Name of slide_____
Magnification_____



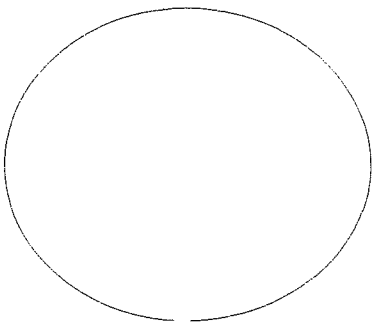
Name of slide_____
Magnification_____



Name of slide_____
Magnification_____



Name of slide_____
Magnification_____



Name: _____ Date: _____

CELLS

The Amazing Microscope

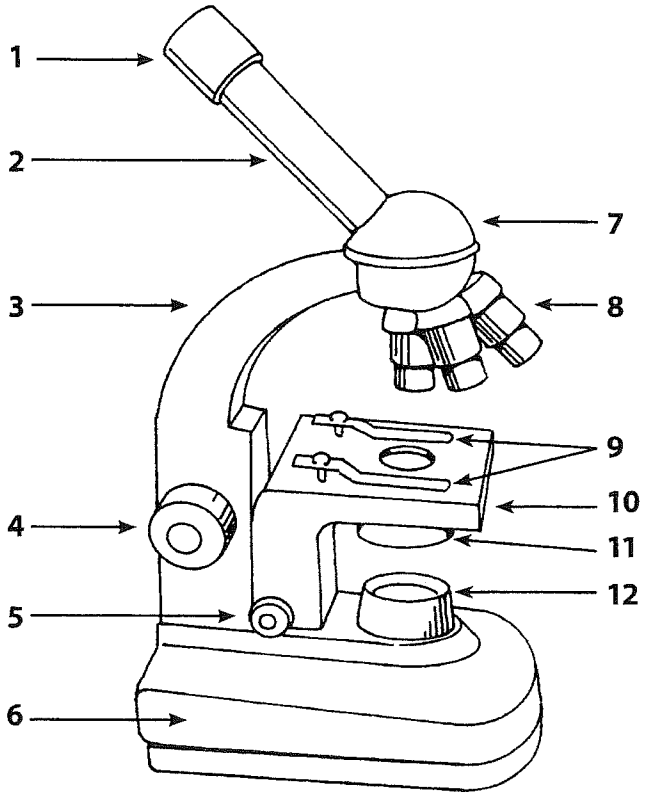
DIAGRAM LABELING

In 1663, an English scientist named **Robert Hooke** placed a piece of cork under a series of lenses. H called the things that he saw through the lens **cells**, because they looked like tiny, rectangular roon A scientist from Denmark named **Anton van Leeuwenhoek** also built his own microscope. When he looked at a sample of pond water through his microscope, he saw tiny organisms swimming in the water. Microscopes have come a long way since the days of those scientists. Scientists now hav powerful microscopes that can help them see things that are extremely small. It is important for scientists to know the parts of a microscope and how to use one to do their work.

DIRECTIONS: Use the words in the word bank to label the diagram of the microscope below.

WORD BANK

fine focus knob	base	stage	nosepiece
arm	light source	coarse focus knob	body tube
objectives	diaphragm	stage clips	ocular (eyepiece)

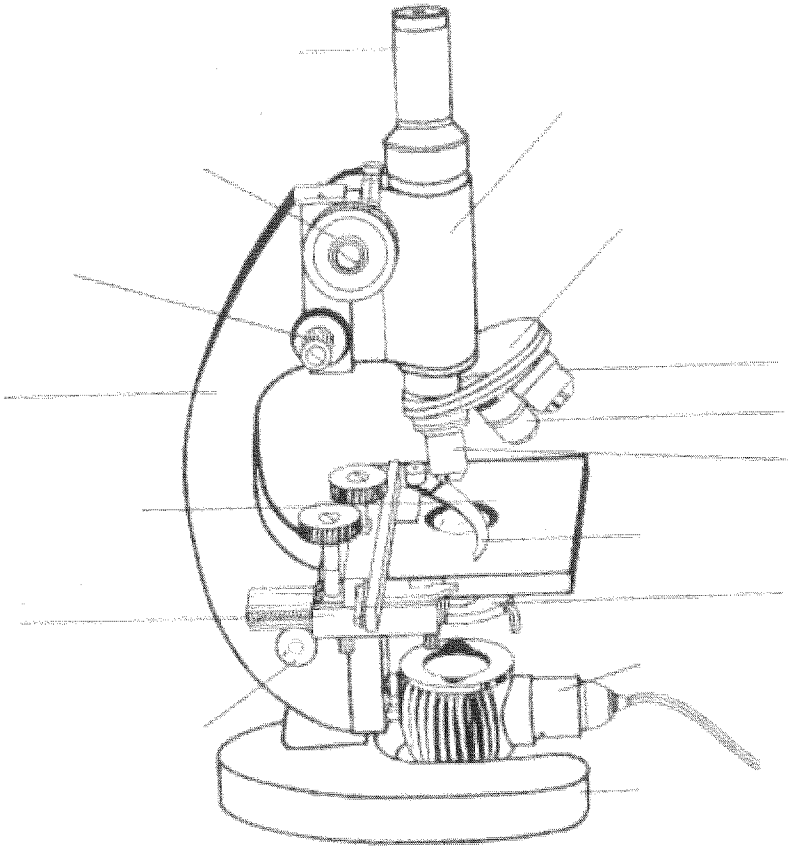


1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____

PARTS OF A MICROSCOPE

Use the word bank to label the parts of a microscope.

eyepiece body tube nosepiece arm
scanner lens stage diaphram base
light source low-power lens high-power lens clip
course adjustment knob fine adjustment knob
inclination joint



Name _____

Microscope Worksheet

1. What is the total magnification for each of the following?

   A) 10X objective, 15X eyepiece _____
   B) 40X objective, 5X eyepiece _____
   C) 4X objective, 10X eyepiece _____
   D) 50X objective, 10X eyepiece _____

2. Draw the following patterns as they would appear when viewed through a compound microscope:

   A) h _____ D) H _____

   B) X _____ E) m _____

   C) 4 _____ F)  _____

3. What is the correct way to hold the microscope when moving it?

   _____

4. What is the field of view? _____

   _____

Use the reading on the back of this worksheet to answer questions 5-7.

5. Who was the first scientist to see cells magnified? _____

6. Why did he call them cells? _____

7. What did the scientist, Leeuwenhoek see when he look at pond water through the microscope?

   _____

8. How many words can you make out of MICROSCOPE MANIA? List them below.

1. _____ holds the objective lenses
2. _____ the lens nearest the stage – the 1st one you use
3. _____ a metal piece that holds the slide in place
4. _____ you carry the microscope with this part
5. _____ the lens you look through
6. _____ the circle of light you see when you look into the microscope.
7. _____ knob used to first find the specimen on the slide
8. _____ the part of the microscope that provides light
9. _____ this controls the amount of light entering the stage
10. _____ the flat part on which you lay the slide

